



Weekly Report

- **Title:** Fortnightly Training Report
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- **Program:** Bachelor of Computer Science
- **Week:** Week 13 and Week 14
- **Dates:** 26 Apr to 7 May 2026
- **Organization/Company Name:** Wadi Makkah Technology Company
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1. Introduction

This report presents the work completed during the thirteenth and fourteenth weeks of cooperative training as part of the Soft-Landing platform project. The implementation phase focused on developing and integrating frontend and backend components, including APIs, database integration, authentication systems, workflow automation, and dashboard interfaces. Thunder Client was used to test APIs and validate request handling, while reusable frontend structures and Magic Pattern were applied to improve consistency and maintainability across the platform. Overall, this experience enhanced practical skills in full-stack development, system integration, and workflow implementation within a real-world environment.

2. Tasks and Responsibilities

Week 13:

Intersect OS Project – Soft Landing Platform

- **Authentication & Authorization System**

During this week, the authentication and authorization system of the Soft-Landing platform was implemented. The work included developing secure user registration and login functionalities using JWT authentication and password hashing techniques.

The registration process supports creating both users and companies simultaneously while validating user data and preventing duplicate accounts. Additionally, JWT tokens were generated to maintain secure authenticated sessions across the platform.

A Role-Based Access Control (RBAC) system was also implemented to manage permissions for different user roles, including Admins, Clients, and Employees. Middleware functions were developed to verify tokens, extract user information, and restrict unauthorized access to protected routes.

In addition, frontend forms and authentication pages were integrated with backend APIs to support login, registration, token handling, and protected routing across the platform interfaces

- **Company & Lifecycle Management**

In addition to authentication, the company management system was implemented. This included creating and updating company profiles, validating ownership permissions, and linking companies to users.

A complete company request lifecycle was also developed to manage the application review process. The lifecycle supports multiple statuses such as UNDER_REVIEW, NEEDS_COMPLETION, APPROVED, and REJECTED.

Submission validation logic was implemented to ensure that companies cannot submit incomplete information. APIs were also integrated with frontend dashboards to display company information, statuses, and workflow updates dynamically.

- **Frontend Integration & API Testing**

During this phase, frontend pages were connected with backend APIs to ensure smooth data flow between the client side and server side. Dynamic forms, task displays, and dashboard components were integrated with API responses.

Thunder Client was used extensively to test API endpoints, validate requests and responses, and verify authentication flows and workflow operations before frontend integration.

In addition, reusable UI structures and Magic Pattern were applied in frontend development to improve consistency, maintainability, and scalability across different interfaces.

Week 14:

- **Workflow Engine Development**

During this week, the Market Entry Engine and workflow automation system were implemented. The workflow engine dynamically generates stages and tasks for each company based on its selected sector.

The implementation included generating company stages automatically after company creation and linking each stage with its corresponding tasks. The system also supports dynamic regeneration of workflow tasks when the company sector changes.



Several APIs were developed to retrieve company stages, company tasks, and workflow progress information. Task management functionality was implemented to support updating task statuses and tracking completion progress.

Frontend dashboard interfaces were also integrated with the workflow APIs to display stages, tasks, progress percentages, and company workflow status dynamically.

- **Progress Calculation & Workflow Logic**

A progress calculation mechanism was implemented using weighted stages and completed tasks. The system calculates the total progress percentage dynamically according to completed tasks and stage weights.

In addition, logic was implemented to determine the company's current stage, next stage, and blocked stages within the workflow process.

Different edge cases were also handled, including:

- Missing stages.
- Missing tasks.
- Sector changes.
- Preventing duplicated tasks and stages.

The workflow engine was fully integrated with the frontend dashboard, backend services, and database successfully.

3. Skills and Knowledge Gained

- Gained experience implementing JWT authentication and RBAC systems.
 - Improved API testing and debugging skills using Thunder Client.
 - Developed workflow automation and progress tracking logic.
 - Improved understanding of reusable frontend structures and Magic Pattern implementation.
 - Enhanced database relationship management and edge-case handling skills.
 - Improved debugging and system integration abilities.
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4. Challenges Faced

1. Workflow Generation Issues

One of the main challenges was dynamically generating workflow stages and tasks based on the selected company sector. Initially, the workflow was generated only when a new company was registered, where the system created stages and tasks linked to the selected sector during the registration process.

However, when the company sector was changed later, the system did not regenerate the workflow correctly. Old stages and tasks remained linked to the company even after selecting a new sector. This caused outdated tasks to continue appearing on the dashboard and resulted in inaccurate progress calculations

because the system relied on both old and newly generated workflow data at the same time.

2. Duplicate Data Inside the Workflow

Another major challenge involved duplicated stages and tasks inside the database when the workflow was regenerated multiple times for the same company, especially after changing the sector or manually modifying workflow stages.

In some cases, the system generated new stages and tasks without removing the previously linked workflow data from the database. As a result, multiple duplicated records for the same stage or task appeared inside the system. This caused repeated tasks to appear on the dashboard and led to incorrect progress calculations because the same tasks were counted multiple times.

The issue also affected the relationships between database tables, as some tasks remained linked to outdated or inactive stages, resulting in inconsistent workflow data and making company status management more difficult.

5. Solutions and Improvements

1. Dynamic Workflow Regeneration

To solve the workflow generation issue, the workflow update logic was redesigned so that the system completely rebuilds the workflow whenever the company sector changes. When a new sector is selected, the system first removes all previously linked company stages and tasks from the database before generating new stages and tasks based on the updated sector.

The workflow generation process was also reorganized so that the system first retrieves only active stages, then fetches the tasks linked to each stage according to the selected sector and finally links them correctly to the company inside the database.

Additional validation logic was implemented to ensure:

- Valid stages exist for the selected sector.
- Each stage contains active tasks.
- Empty stages or tasks are not generated.
- Database insert operations complete successfully before moving to the next step.

If invalid data or workflow generation errors occur, the system now stops the process completely and returns proper error messages instead of inserting incomplete or inconsistent workflow data into the system.

This solution ensured that workflow stages and tasks always remain synchronized with the company's current sector while improving the accuracy of workflow progress tracking.



2. Duplicate Prevention and Edge Case Handling

To prevent duplicated stages and tasks, additional validation logic was added before creating any new workflow records. The system now checks whether company workflow data already exists before generating new stages and tasks.

The regeneration process was also updated so that all old workflow records linked to the company are deleted before inserting new records, preventing duplicated or outdated data from remaining inside the database.

Several edge cases were also handled safely, including:

- Missing stages for a selected sector.
- Missing tasks inside a stage.
- Inactive stages.
- Multiple sector changes.
- Partial workflow generation failures.

In addition, clear error messages were added to stop invalid workflow generation processes and prevent incomplete workflow structures from being created.

These improvements helped maintain database consistency, ensured correct dashboard data display, and improved the accuracy of workflow progress calculations.

6. Conclusion

In conclusion, the thirteenth and fourteenth weeks represented an advanced practical stage in the cooperative training journey, where the work progressed into building an integrated platform that combines both frontend and backend components within a real-world environment. This experience contributed to enhancing the ability to connect different system components and handle technical challenges related to API integration, database management, and workflow automation.

In addition, working on authentication and authorization systems, as well as the workflow engine, provided a deeper understanding of how scalable digital systems are developed while maintaining system stability.

Despite the significant progress achieved, two major phases are still under development, including the document and licensing management system, as well as the completion of dashboards and notification systems within the platform.

End of Report
